



M1.2 Unit Overview

Linear Equations and Inequalities

Students revisit strategies to determine the **solutions** to one-variable linear equations and extend their knowledge to make sense of solving multi-variable linear equations. They create **equivalent equations** as they solve linear equations and interpret the different types of solutions such as **infinitely many solutions** and **no solutions** in context. They connect graphs, tables, and equations to the situations they represent and interpret key features such as the **x-intercept** and **y-intercept**. Students represent the same linear equation in **point-slope form**, **standard form** or **slope-intercept form**. They determine the **solution set** to an inequality algebraically and they use technology tools to strategically find the graphical **solution region** to inequalities. They investigate real situations and model their **constraints** using equations and inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations, NS: Financial Literacy

Explore: Planning for Homecoming

(Optional) Launch the unit with a non-routine task to investigate the question, "How can different conditions affect preparation for Homecoming?"

Sub-Unit 1: One-Variable Equations (Lessons 1–3)

- Solve linear equations with one variable, including equations with no solution or infinitely many solutions.
- Language Goal:** Describe strategies for solving one-variable linear equations, including justifying the number of solutions a one-variable equation will have.

Sub-Unit 2: Multi-Variable Equations (Lessons 4–8 + Practice Day + Quiz)

- Solve multi-variable equations for a given variable.
- Create equations, graphs, and tables to represent linear relationships and interpret their solutions in context.
- Language Goal:** Describe connections between equations, tables, descriptions, and graphs of linear situations. (*Writing and Speaking*)

Sub-Unit 3: One-Variable and Two-Variable Inequalities (Lessons 9–17 + Practice Day)

- Determine solutions to an inequality algebraically and graphically.
- Write inequalities in one and two variables to represent constraints.
- Language Goal:** Interpret the solutions to one- and two-variable inequalities in context. (*Reading, Writing, and Speaking*)

Contextual Vocabulary

budget, committee, homecoming dance, taxi, seating capacity, standing capacity, subway, fair, trampoline, bookcase, screws, beads, bracelet, flower bed, vegetable, venue, dam, distribution

Standards

Addressing	A-CED.1	A-CED.2	A-CED.3	A-CED.4	A-REI.1	A-REI.10	A-REI.12	A-REI.3	A-REI.3.1	A-SSE.1.a	F-IF.4	F-IF.7	F-IF.7a	F-LE.5
	N-Q.1	N-Q.2	N-Q.3	SMP.1	SMP.2	SMP.3	SMP.4	SMP.6	SMP.7	SMP.8				

Vocabulary

- absolute value
- boundary line
- compound inequality
- constraint
- equivalent equations
- infinitely many solutions
- no solution
- point-slope form
- slope-intercept form
- solution
- solution region
- solution set
- standard form (of a linear equation)
- x-intercept
- y-intercept

Review Vocabulary

- distributive property
- equation
- inverse operations
- linear relationship
- slope
- substitute
- variable



M1.2 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete using a scientific calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Unit Rates in the World (Grade 7), Linear Equations, Multiple Representations of Functions (Grade 8)

Standards

Assessing

6.EE.8

7.EE.1

7.EE.4

8.EE.5

8.EE.7

8.EE.7.a

8.EE.7.b

SMP.2

SMP.4

SMP.6

SMP.7



Explore: Planning for Homecoming

How can different conditions affect preparation for homecoming?

OPTIONAL

This lesson is optional because the context extends beyond the scope of the standards.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Idea: NS: Financial Literacy

Today's Goals

- Use conditions from real-world contexts to maximize value.
- Determine which choices meet given conditions in a real-world context.
- **Language Goal:** Explain a method for maximizing value while meeting the conditions in a real-world context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Materials

Activity Cards (one set per group, pre-cut), Activity Sheet, scientific calculators (optional)

Standards

Building On 7.EE.3 Addressing N-Q.1 SMP.1 SMP.2 Building Toward A-CED.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.5



Working Backwards

Lesson 1: Solving Equations With Inverse Operations

Let's solve equations by working backwards.

📧 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use interactive activities that offer Responsive Feedback to support students in developing fluency.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Use inverse operations to solve an equation.
- **Language Goal:** Describe strategies for solving multi-step equations. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

Building On 6.EE.5 8.EE.7 Addressing A-REI.3 SMP.1 SMP.2 SMP.7 Building Toward A-CED.4 A-REI.3

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- solution

Review Vocabulary

- equation
- inverse operations

Instructional Routines

- Compare and Connect (MLR7)



Solving Strategies

Lesson 2: More Solving With One-Variable Equations

Let's explore strategies for solving equations.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve one-variable linear equations.
- Analyze others' reasoning when solving linear equations.
- Language Goal:** Justify that each step in solving a linear equation creates an *equivalent equation* with the same solution. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 8.EE.7.b Addressing A-REI.1 A-REI.3 SMP.3 SMP.6 SMP.7 Building Toward A-CED.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- equivalent equations

Review Vocabulary

- distributive property

Instructional Routines

- Collect and Display (MLR2)
- Think-Pair-Share
- Notice and Wonder
- Critique, Correct, Clarify (MLR3)



Same Position

Lesson 3: No Solution and Infinitely Many Solutions

Let's explore how many solutions are possible for a one-variable equation.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to develop a conceptual understanding of equations with no solution and infinitely many solutions.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations

Today's Goals

- Observe that dividing by a variable does not always create equivalent equations.
- Language Goal:** Justify whether a one-variable equation has one solution, *no solution*, or *infinitely many solutions*. (*Listening, Speaking, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 8.EE.7.a Addressing A-CED.1 A-REI.3 SMP.1 SMP.3 SMP.7 Building Toward A-CED.1

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6 ELD.PI.9-10.9

Vocabulary

- infinitely many solutions
- no solution

Instructional Routines

- Tell a Story
- Decide and Defend
- Compare and Connect (MLR7)



Subway Seats

Lesson 4: Representing Situations With Two-Variable Equations

Let's explore what different forms of linear equations reveal about a situation.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to develop conceptual understanding of what their solutions mean in context.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Identify equivalent forms of linear equations and understand that different forms of a linear equation can be useful for different purposes.
- Represent constraints with two-variable equations and interpret their solutions.
- Language Goal:** Interpret different forms of a two-variable linear equation in context. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

constraint

Standards

Building On A-REI.3 Addressing A-CED.3 A-CED.4 SMP.2 SMP.6 SMP.8 Building Toward A-CED.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5

Instructional Routines

- Which One Doesn't Belong?
- Think-Pair-Share



Various Variables

Lesson 5: Solving Multi-Variable Equations

Let's rearrange equations with multiple variables.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Rearrange multi-variable equations to highlight a variable of interest.
- Solve multi-variable equations for a variable of interest.
- Language Goal:** Explain how solving for a variable is similar to rearranging an equation to isolate a variable of interest. (*Speaking and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

chips or coins (as needed); highlighters (as needed); Show What You Know Sheet

Standards

Building On 8.EE.7 Addressing A-CED.4 A-REI.3 SMP.8 Building Toward A-REI.3 N-Q.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Review Vocabulary

- substitute
- variable

Instructional Routines

- Think-Pair-Share
- Compare and Connect (MLR7)



Shelley the Snail

Lesson 6: Connecting Graphs and Linear Equations

Let's connect graphs, tables, and equations to the situations they represent.

📧 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students make connections across representations with Responsive Feedback.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations

Today's Goals

- Understand that the graph of a linear equation represents all the pairs of values that are solutions to that equation.
- Solve an equation in standard form for one of its variables and connect the new equation to its graph.
- **Language Goal:** Discuss how the graph of a linear equation can be written in different but equivalent forms. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.EE.5 8.F.3 Addressing A-CED.2 A-CED.3 A-CED.4 A-REI.10 F-IF.4 F-IF.7a N-Q.2 SMP.1 SMP.2

Building Toward A-CED.2 A-CED.3 A-CED.4

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- x-intercept
- y-intercept

Review Vocabulary

- linear relationship

Instructional Routines

- Critique, Correct, Clarify (MLR3)



Equations of Lines

Lesson 7: Introducing Point-Slope Form

Let's explore what information is needed to determine a point-slope form equation of a line.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Systems of Equations

Today's Goals

- Generalize that a line can be represented by an equation in *point-slope form*.
- Write the equation of a line in *point-slope form*.
- Language Goal:** Describe the properties of a line when given its equation in *point-slope form*. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 2 Cards; Show What You Know Sheet

Standards

Building On 8.EE.6 Addressing A-CED.2 A-SSE.1.a SMP.3 SMP.6 SMP.7 Building Toward G-GPE.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.5

Vocabulary

- point-slope form

Review Vocabulary

- slope

Instructional Routines

- Think-Pair-Share
- Critique, Correct, Clarify (MLR3)
- Decide and Defend



Five Representations

Lesson 8: Linear Relationships in Equations, Tables, and Graphs

Let's explore different ways to represent linear relationships.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations

Today's Goals

- Write two linear equations to represent the same situation: one in *standard form* and one in *slope-intercept form*.
- Language Goal:** Describe connections between equations, tables, descriptions, and graphs of linear situations. (*Writing, Speaking, and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 1 Cards (one per pair, pre-cut); coloring tools (as needed); group-sized whiteboards or chart paper (optional); Show What You Know Sheet

Standards

Building On 8.EE.7 8.F.3 Addressing A-CED.2 A-CED.4 F-IF.4 F-IF.7 F-IF.7a F-LE.5 N-Q.1 SMP.1 Building Toward A-CED.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- standard form (of a linear equation)
- slope-intercept form



M1.2 Practice Day 1



Students practice the concepts, skills, and strategies developed in Lessons 1–8. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz. https://learning.amplify.com/m/3ae1487729f0b46e/original/MATH_M1_Overview_CA_27_0090.png

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations

Additional Materials

Option 1: Stations

Task Cards (two sets for the whole class)

Option 2: Group Problems

Task Cards (one set per group)

Standards

Addressing [A-CED.2](#) [A-CED.3](#) [A-CED.4](#) [A-REI.1](#) [A-REI.10](#) [A-REI.3](#) [A-SSE.1.a](#) [SMP.2](#) [SMP.3](#) [SMP.7](#)



M1.2 Sub-Unit Quiz

Learn about your students' understanding of the concepts and skills so far in this unit.
This assessment is intended for students to complete using a scientific calculator.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations
Standards

Assessing

A-CED.2

A-CED.4

A-REI.1

A-REI.10

A-REI.3

SMP.2

SMP.3

SMP.7



Pizza Delivery

Lesson 9: Representing Situations With One-Variable Inequalities

Let's write inequalities to represent constraints.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to develop conceptual understanding of how to write inequalities to represent constraints.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations, NS: Financial Literacy

Today's Goals

- Understand that solutions to an inequality are values that make the inequality true.
- Write one-variable inequalities that represent *constraints*.
- Language Goal:** Describe and interpret solutions to one-variable inequalities in context. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 6.EE.8 7.EE.4 Addressing A-CED.1 A-CED.3 N-Q.1 SMP.1 SMP.2 SMP.3 Building Toward A-REI.3

Language Standards

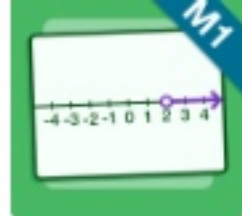
ELD.PI.9-10.10 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6 ELD.PI.9-10.9

Vocabulary

- constraint

Instructional Routines

- Tell a Story
- Compare and Connect (MLR7)



Graphing Inequalities

Lesson 10: Inequalities on the Number Line

Let's represent solutions to inequalities on a number line.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Teachers can use the dashboard's Teacher View to quickly access student thinking and identify students who need additional support.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve one-variable inequalities by reasoning.
- Graph solutions to a one-variable inequality on a number line.
- Language Goal:** Describe how to represent the solutions to a one-variable inequality on a number line. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 7.EE.4 Addressing A-REI.3 SMP.1 SMP.2 SMP.3 Building Toward A-REI.12

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- solution region
- solution set

Review Vocabulary

- boundary point

Instructional Routines

- Critique, Correct, Clarify (MLR3)
- Decide and Defend



Solutions and Sheep

Lesson 11: Solving One-Variable Inequalities

Let's make connections between solving one-variable equations and solving inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Repeated Challenges offer Responsive Feedback to support students in developing fluency.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve a one-variable linear inequality by solving its corresponding equation and testing values.
- Language Goal:** Describe a strategy for solving and graphing the solutions to an inequality. (*Listening, Speaking, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

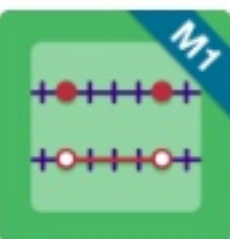
Building On 7.EE.4 8.EE.7 Addressing A-REI.3 SMP.1 SMP.3 Building Toward A-REI.12

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PII.9-10.4

Instructional Routines

- Notice and Wonder
- Decide and Defend



Absolute Value Solutions

Lesson 12: Solving Absolute Value Equations and Inequalities

Let's solve absolute value equations and inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Repeated Challenges offer Responsive Feedback to support students' development of strategies for computation and problem solving.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations

Today's Goals

- Solve one-variable absolute value equations and inequalities.
- Graph solutions to absolute value equations and inequalities on a number line.
- Language Goal:** Explain the meaning of absolute value equations and inequalities. (Writing, Speaking, and Listening)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 6.NS.7.c A-REI.3 Addressing A-CED.1 A-REI.3.1 SMP.2 SMP.7 Building Toward A-CED.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- absolute value

Instructional Routines

- Compare and Connect (MLR7)
- Think-Pair-Share



All Sorts of Screws

Lesson 13: Creating and Solving Absolute Value Equations and Inequalities

Let's create and solve absolute value equations and inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to build conceptual understanding of how to write absolute value equations and inequalities for problems in context.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations

Today's Goals

- Create absolute value inequalities to solve problems.
- Solve one-variable absolute value equations and inequalities.
- **Language Goal:** Explain the meaning of absolute value equations and inequalities, and their solutions in context. (*Listening, Speaking, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Standards

Building On 6.NS.7 6.NS.7.c A-REI.3 Addressing A-CED.1 A-REI.3.1 SMP.1 SMP.3 SMP.7 Building Toward A-REI.10

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- compound inequality

Instructional Routines

- Notice and Wonder
- Decide and Defend



Bracelet Budgets

Lesson 14: Introduction to Two-Variable Inequalities

Let's explore solutions to two-variable inequalities graphically and symbolically.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to develop conceptual understanding of what solutions to a two-variable inequality look like in multiple representations.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations, NS: Financial Literacy

Today's Goals

- Understand that the solutions to a two-variable inequality are points that make the inequality true.
- Language Goal:** Interpret two-variable inequalities that represent constraints in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

coloring tools (optional)

Standards

Building On [6.EE.8](#) [8.EE.5](#) Addressing [A-CED.3](#) [A-REI.10](#) [N-Q.2](#) [SMP.1](#) [SMP.2](#) [SMP.3](#) [SMP.4](#) Building Toward [A-REI.12](#)

Language Standards

[ELD.PI.9-10.1](#) [ELD.PI.9-10.11](#) [ELD.PI.9-10.12](#) [ELD.PI.9-10.2](#) [ELD.PI.9-10.3](#) [ELD.PI.9-10.5](#) [ELD.PI.9-10.6](#)

Instructional Routines

- Notice and Wonder
- Think-Pair-Share
- Compare and Connect (MLR7)



All of the Solutions

Lesson 15: Graphing Solutions to Two-Variable Inequalities

Let's represent all of the solutions to two-variable inequalities graphically.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to make connections across multiple representations and revise their graphs of solutions to two-variable inequalities.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Understand that the solutions to a two-variable linear inequality are represented on a graph as a half-plane (excluding the boundary in the case of a strict inequality).
- Graph the solutions to a two-variable linear inequality given the graph of its corresponding equation.
- Language Goal:** Describe a strategy for graphing the solutions to a two-variable linear inequality. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Additional Vocabulary

half-plane

Standards

Building On A-REI.10 Addressing A-REI.12 F-IF.7 SMP.3 SMP.6 SMP.8 Building Toward A-REI.12

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- boundary line

Review Vocabulary

- solution

Instructional Routines

- Decide and Defend
- Critique, Correct, Clarify (MLR3)
- Think-Pair-Share



Charity Concerts

Lesson 16: Graphing Two-Variable Inequalities in Context

Let's represent constraints by graphing two-variable inequalities in context.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations, NS: Financial Literacy

Instructional Routines

- Three Reads (MLR6)

Today's Goals

- Graph the solutions to a two-variable linear inequality and interpret its solutions in context.
- Write a two-variable inequality to represent a situation in context.
- **Language Goal:** Interpret what an inequality and its solutions mean in context. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

scientific calculators (optional); coloring tools (optional), Show What You Know Sheet

Standards

Building On A-REI.3 Addressing A-CED.3 A-REI.10 A-REI.12 F-IF.7 N-Q.1 N-Q.2 N-Q.3 SMP.1 SMP.4 SMP.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.6 ELD.PI.9-10.8

Water Way

Lesson 17: Using Two-Variable Inequalities to Make Decisions

Let's use two-variable inequalities to make decisions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations

Instructional Routines

- Notice and Wonder

Today's Goals

- Model real-world constraints and solutions using two-variable linear inequalities.
- **Language Goal:** Analyze two-variable linear inequalities in order to make sense of an issue in society. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

CA EP&C Principle 2

Additional Materials

Activity 2 Cards (one set per group, pre-cut), Show What You Know Sheet

Standards

Building On A-CED.2 A-REI.3 Addressing A-CED.3 A-REI.12 F-IF.7 SMP.1 SMP.3 SMP.4 Building Toward A-CED.3 A-REI.12

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.5 ELD.PII.9-10.4



M1.2 Practice Day 2



Students practice the concepts, skills, and strategies developed in Lessons 1–17. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

🗨️ Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations, NS: Financial Literacy

Additional Materials

scientific calculators (optional)

Standards

Addressing A-CED.3 A-CED.4 A-REI.10 A-REI.12 A-REI.3 SMP.2

Language Standards

ELD.PI.9-10.5 ELD.PI.9-10.6



M1.2 Performance Task

Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, Systems of Equations, NS: Financial Fluency

Standards

Assessing A-CED.2 A-CED.3 A-CED.4 A-REI.10 A-REI.12 SMP.2 SMP.3 SMP.4



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

🗉 Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.